Rui Zhou

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SUMMARY

Computational materials PhD candidate (expected May 2027; available from June 2027) combining DFT with active-learning ML interatomic potentials to study interfaces, phase transitions and ion transport in solid electrolytes. At Sunwoda, trained ML force fields and screened cathode dopants with high-throughput DFT; at Seagate, built GNN chemical-property models. 4 granted patents, 4 released software applications, 13 peer-reviewed publications (5 first-author).

EXPERIENCE

PhD Researcher - Computational Materials and ML Potentials at Iowa State University *Ames, Iowa | Jan 2023 - Present*

- Train ML interatomic potentials through an active-learning workflow with reference data from multiple electronic-structure methods, including DFT, constrained DFT for excited-state semiconductors and MP2 for combustion chemistry.
- Resolved atomic-scale interphase-formation mechanisms at Li/glassy-sulfide electrolyte interfaces using ML potentials trained on first-principles data with QEq charge equilibration; first-author paper, J. Mater. Chem. A 2026.
- Showed that anion topology and cation-anion coupling govern phase transitions and ionic transport in closo-hydroborate electrolytes using ML-force-field MD; first-author papers, ACS Appl. Energy Mater. 2026 and ACS Electrochemistry 2025.
- Linked glass structure to Li-ion conductivity in lithium sulfide glasses with ML-force-field simulations (first author, ACS Appl. Mater. Interfaces 2024); first-authored a Chem. Commun. 2025 article on data-driven atomistic modeling of solid electrolytes.
- Apply Bayesian optimization to glassy solid-electrolyte composition searches; explore graph neural networks and diffusion models for property prediction and glassy-structure generation.
- Contributed to ML-force-field studies of ZnS and CdTe photoplasticity, Mg-Bi alloy dislocation slip and high-entropy carbide deformation; coauthor, ACS Mater. Lett. 2025, Int. J. Mech. Sci. 2025 and Phys. Rev. Mater. 2026.
- Build Python, NumPy, PyTorch and MLX workflows on Slurm/HPC that automate job preparation, execution, post-processing and model validation; coauthored the dpdata paper on atomistic ML datasets (J. Chem. Inf. Model. 2025).
- Developed and released NAIIVE, a Swift/Metal application for 2D and spatial 3D visualization of atomistic simulations and MLIP datasets on macOS, visionOS and iOS.

Advanced Data Storage Material Discovery Intern at Seagate Technology *Normandale, Minnesota | May 2026 - Aug 2026*

- Developed an end-to-end machine-learning workflow for thin-film ellipsometry data analysis and materials characterization.
- Built graph-neural-network and RDKit-derived descriptor models for chemical-property prediction.
- Applied molecular dynamics to study atomic structure and behavior in reader-head coating materials.

Simulation Engineer at Sunwoda Electric Vehicle Battery Co., Ltd. *Shenzhen, China | Dec 2020 - Dec 2022*

- Trained and validated ML force fields for cathode materials, automating generation of DFT reference structures.
- Performed high-throughput DFT and force-field dopant screening for NCM- and LFMP-based cathodes; contributed to granted patent CN114744187B on a ternary cathode material.
- Developed scikit-learn and symbolic-regression models for battery power mapping and swelling-force prediction.
- Built DEM calendaring models to optimize particle-size distribution and mixing ratio, and automated parameter identification for DEM, P2D/ECM and FEM models; granted patents CN114169233B, CN115579085B, CN115392123B.

EDUCATION

Iowa State University *Ames, Iowa | Jan 2023 - Expected May 2027*
PhD in Materials Science and Engineering

Washington University in St. Louis *St. Louis, Missouri | Aug 2018 - Dec 2019*
M.S. in Materials Science and Engineering

Changchun University of Science and Technology *Changchun, China | Sep 2014 - Jun 2018*
B.S. in Materials Science and Engineering

SKILLS

Modeling Methods: DFT, Molecular Dynamics, ML Interatomic Potentials, Active Learning, Bayesian Optimization, GNNs, QEq

Selected Tools: Python, NumPy, PyTorch, MLX, scikit-learn, RDKit, VASP, Quantum ESPRESSO, GAMESS, LAMMPS, ASE, Slurm/HPC, Swift/Metal